

ANKUR KOHLI

SOFTWARE ENGINEER

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ANKUR KOHLI - PORTFOLIO Genoa, Italy Indian

SOFTWARE ENGINEER | FULL-STACK & AI/GENAI

PROFESSIONAL SUMMARY

Software Engineer experienced in designing and developing scalable software systems across frontend and backend environments for the Italian MediaTech industry, Japanese clients, and other international engagements. Experience spans backend services, API driven and real time applications, distributed systems, and the integration of AI and LLM capabilities into software products. Contributes to system design, implementation, integration, testing, deployment, and optimization, with an emphasis on performance, maintainability, and sound software architecture.

PROFESSIONAL EXPERIENCE

▸ **Freelance Software Engineer & Consultant** *December 2025 - Present*

GINI Technologies US, Remote

- Deliver end-to-end full stack features across frontend and backend for international clients, collaborating with stakeholders on requirements, implementation, testing, deployment, and maintenance.
- Develop and integrate LLM based services for classification, summarization, document retrieval, and automated response generation.
- Design serverless AI/ML and GenAI pipelines across AWS, Azure, and GCP, integrating AI vision services (Rekognition, Cloud Vision, Azure AI Vision), GenAI models (Vertex AI Gemini, Azure OpenAI, Amazon Bedrock), NoSQL databases, REST APIs, IAM, and event driven automation for image analysis, content moderation, and caption generation.
- Designed, implemented, and deployed a full stack RAG application using FastAPI, ReactJs, and PostgreSQL/pgvector, enabling PDF question answering with source citations through an MCP server compatible with AI clients such as Claude Desktop.

▸ **Software Engineer** *April 2025 - October 2025*

NTT DATA Italia Italy

- Developed software for frontend and backend components for enterprise MediaTech solutions in the Italian market, working within multithreaded and distributed system architectures.
- Designed and integrated communication components for low latency data exchange between system modules in real time applications.
- Led the deployment of an autonomous navigation system for an international client, from technology assessment through physical robot validation, translating research requirements into a repeatable operational workflow.
- Designed and implemented a testing framework for integration testing, regression checks, and runtime verification, standardizing release processes across the team.

▸ **Software Engineering Thesis Project** *June 2023 - February 2024*

NTT DATA Italia Italy

- Designed a ROS2 based software architecture for autonomous quadrupedal and mobile robots, integrating perception, localization, mapping, and navigation components.
- Developed Dockerized SLAM and navigation environments using ROS2, supporting reproducible development and experimentation.
- Implemented autonomous exploration and navigation workflows using SLAM, localization, path planning, and navigation algorithms.
- Evaluated robot behavior in simulation and on physical platforms, identifying navigation and integration issues and refining the system through iterative testing.

PROFESSIONAL SKILLS

TECHNICAL SKILLS

Programming Languages:
Python, C/C++, Java, Dart, JavaScript

Frontend Technologies:
HTML5, CSS3, ReactJS, Bootstrap, Tailwind CSS

AI, GenAI & LLM:
LLMs, RAG, MCP, Gemini/OpenAI API, AI Vision Services (AWS, GCP, Azure)

Backend & Distributed Systems:
FastAPI, Spring Boot, REST APIs, WebSocket Services, JSON, SQL (MySQL, PostgreSQL), pgvector, NoSQL

Software & Tools:
AWS, Microsoft Azure, Google Cloud Platform (GCP), Ubuntu/Linux, Docker, Git/GitLab, Flutter

SOFT SKILLS

Leadership & Planning:
Strategic Planner, Project Management, Agile Delivery, Stakeholder Communication

Innovation & Problem Solving:
Structured Problem Analysis, Technology Assessment, Change Management Support

Teamwork & Collaboration:
Reliable, Organized, Cross functional Collaboration

CERTIFICATIONS

- **AWS Training and Certification:** Generative AI Developer – Professional: Domain 1 Review *October 2026*
- **Hugging Face:** Fundamentals of Agents *October 2026*
- **Anthropic:** Model Context Protocol: Advanced Topics *September 2026*
- **OpenAI:** Apply AI at Work Pathway Completion *September 2026*
- **Anthropic:** AI Fluency: Framework & Foundations *August 2026*
- **Anthropic:** Claude Platform 101 *August 2026*
- **Microsoft:** Github Copilot Fundamentals *July 2026*
- **Google Cloud:** Prompt Design in Vertex AI *February 2026*
- **Microsoft:** Azure Fundamentals *November 2025*
- **Google Cloud:** Introduction to Large Language Models *January 2025*

EDUCATION

- **M.Sc Computer Engineering (Specialization: Robotics Engineering)** *September 2021 - March 2024*
  University of Genoa Italy
- **B.Tech Mechatronics Engineering** *July 2016 - June 2020*
  University of Petroleum & Energy Studies India

PROJECTS

- **Software Development Portfolio**
  - **Research Assistant Pro | Full-Stack Chrome Extension & Reactive AI Backend:** Built a Chrome extension using Spring Boot and Gemini API for AI summarization, Q&A, notes, and referencing. [GitHub]
  - **ResumeAI: Next-Gen Career Intelligence Platform:** Designed a cost effective AI resume analyzer using React.js, Tailwind CSS, and Puter.js for ATS scoring, JD matching, and skill gap analysis. [GitHub]
- **Artificial Intelligence Portfolio**
  - **Automated Warehouse Scenario Using PDDL 2.1:** AI planning system using PDDL 2.1, priority queues, and plan graphs to optimize order management, storage allocation, and logistics flow. [GitHub]
  - **Task and Motion Planning for Robotics in Coffee Shop Scenario:** Integrated task and motion planning for robotic navigation using PDDL, state space graphs, BFS, heuristic search, and Euclidean distance. [GitHub]
- **Robotics Portfolio**
  - **Software Architecture for Mobile Robot Control:** ROS based robot controller using Dijkstra's algorithm for graph based routing and autonomous path planning in Python. [GitHub]

LANGUAGE

- **Italian:** Functional Proficiency
- **English:** Proficient